

Prompt 1: To create a date table

create a date table that has the dates from 03-01-2025 (March 1st, 2025) to 05-31-2025 (May 31st, 2025)

Prompt 2: To create an exchange rate table

Create an exchange_rate table for the period from March 1st, 2025, to May 17th, 2025. Use the following API request sample to fetch historical exchange rates:

```
response = requests.get(f'https://openexchangerates.org/api/historical/2024-10-21.json?app_id=<Replace your app id here>&symbols=INR')
```

Replace the date for each in the URL with each date within the specified range.

Prompt 3: Data cleaning and summarizing required data in one table (fact_summary)

Create a Python code that:

1. Load data from:

- fact_order_line table
- dim_products sheet (skip first row, use second row as headers)
- dim_customers sheet (skip first row, use second row as headers)
- Exchange Rate sheet (skip first row, use second row as headers)

2. Cleans the data:

- Convert product_id and customer_id to numeric
- Strip whitespace from IDs
- Remove rows with NULL IDs
- Convert IDs to integers
- Convert dates to datetime

3. Merges the tables:

- Orders with products using product_id
- Result with customers using customer_id
- Result with exchange rates using order_placement_date

4. Calculates total amounts:

- For USD currency: price_USD * USD_INR_Rate * order_qty
- For INR currency: price_INR * order_qty

5. Clean final output:

- Drop intermediate columns: price_USD, price_INR, currency, Date, USD_INR_Rate
- Keep only essential columns: order details, IDs, dates, quantities, delivery information, and total_amount (in INR)

Prompt 4: Creating Business KPIs

Create the following KPIs:

1. Total Order Lines
2. Line Fill Rate
3. Volume Fill Rate
4. Total Orders
5. On Time Delivery %
6. In Full Delivery %
7. On Time In Full %